

Date	30 September 2026
Team ID	SWTID-2026-7220
Project Name	import data using transform maps(spreadsheet)
Maximum Marks	2 Marks

Project Design Phase Proposed Solution Template

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Manually importing bulk data from spreadsheets (.csv, .xlsx) into relational databases is highly error-prone, time-consuming, and lacks a standardized method to map varying column headers to fixed database schema fields, leading to data corruption and system crashes.
2.	Idea / Solution description	Develop a robust web-based application that allows users to upload spreadsheets and visually map the spreadsheet columns to target database fields using "Transform Maps." The system will validate data types before ingestion, and allow users to save these mappings as reusable templates for future data imports, fully automating the extraction and loading process.
3.	Novelty/ Uniqueness	The solution introduces intelligent auto-mapping capabilities (using string matching algorithms) to suggest column pairs. Additionally, it separates the UI thread from the data ingestion thread by utilizing a background message queue (e.g., RabbitMQ), ensuring the application remains responsive even when processing files with hundreds of thousands of rows.
4.	Social Impact / Customer Satisfaction	By eliminating tedious manual data entry, the system significantly boosts productivity and morale for data entry operators and administrators. It ensures high data integrity and reliability, which leads to better business decision-making and heightened customer satisfaction due to zero data-loss imports.
5.	Business Model (Revenue Model)	A SaaS (Software as a Service) subscription model. A